

FRAUD DATA ANALYTICS PROJECT

Scenario-Driven Detection of Unauthorized Account Usage

Description:

This project applies a scenario-driven fraud data analytics methodology to identify transaction patterns consistent with unauthorized use of customer financial accounts. The analysis focuses on behavioural and temporal anomalies associated with account takeover and dormant account exploitation.

Name: Arisha Shtypel

Field: Fraud Analytics / Financial Crime

Tools: SQL (MySQL)

Dataset: Simulated Bank Transactions Dataset (Kaggle)

Year: 2026

OVERVIEW

This project implements a scenario-driven fraud detection framework based on fraud data analytics methodology applied to transactional banking data.

DATA

Project uses Bank Transactions Dataset for Fraud Detection from Kaggle <https://www.kaggle.com/datasets/thuandao/bank-transactions-dataset-for-fraud-detection>

The dataset consists of 50,000 bank transaction records that simulate real-world banking activity.

Data availability:

1. The following variables are available:

TransactionID → Unique identifier for each transaction.

AccountID → Unique identifier for the bank account involved in the transaction.

TransactionAmount → Amount of money transferred in the transaction.

TransactionDate → Date when the transaction occurred.

TransactionType → Type of transaction (e.g., withdrawal, deposit, transfer, payment).

TransactionDuration → Time taken to complete the transaction (min).

AccountBalance → Account balance after the transaction.

Channel → Channel used to perform the transaction (ATM, mobile banking, online banking, etc.).

Location → Geographic location where the transaction was initiated.

DeviceID → Identifier of the device used for the transaction.

IP Address → IP address associated with the transaction session.

MerchantID → Identifier of the merchant involved in the transaction (if applicable).

CustomerAge → Age of the account holder.

CustomerOccupation → Occupation category of the customer.

LoginAttempts → Number of login attempts recorded before completing the transaction.

2. There are no missing values in any of the fields
3. TransactionDate variable is only partly usable, only some have the timestamp, others only have the date.

Reliability:

TransactionID is a control number that follows a sequential pattern based on transaction date. To check reliability, tests were conducted to establish if there are 1) missing transaction_id 2) duplicate transaction_id 3) out of order transaction_id. The tests proved that none are the case.

Usability:

Data is fully usable, with an exception of transasction_date field that had to be recoded from text to datetime format and standardised.

Formatting:

Column names were converted to snake case. Date field was converted from text to datetime and can now be used in meaningful analysis.

SCOPE

Primary fraud category: Asset misappropriation

Secondary fraud category: Theft of monetary funds

Inherent fraud scheme: Unauthorized control and use of a customer financial account to execute transactions resulting in the misappropriation of funds.

Time period: 5 years.

The fraud data analytics scope focuses on identifying transaction patterns consistent with unauthorized account usage rather than confirming fraudulent activity

OBJECTIVE

Determine whether customer financial accounts exhibit transaction patterns consistent with unauthorized use resulting in the misappropriation of funds.

METHODOLOGY

This project applies a scenario-driven fraud data analytics approach. The methodology is based on identifying specific fraud scenarios and translating them into observable data patterns.

Each fraud scenario is decomposed into:

- the person committing the fraud
- the entity involved
- the fraudulent action
- the associated concealment strategies

Concealment strategies are translated into red flags, which are then operationalised through data interrogation tests. The objective of the analysis is not to identify confirmed fraud, but to isolate transactions that exhibit patterns consistent with defined fraud scenarios.

FRAUD RISK ASSESSMENT

Two fraud scenarios were derived from the inherent scheme (there are more scenarios that can be explored, however for the purpose of the current project two were deemed enough).

Scenario 1: An external fraudster gains unauthorized access to an active customer account and executes multiple high-frequency transactions within a short time window to extract funds before detection.

Scenario 2: An external fraudster identifies and gains access to a dormant or inactive customer financial account and initiates transaction activity after a prolonged period of inactivity to extract funds, resulting in financial loss.

These scenarios represent potential mechanisms through which unauthorized account use may occur.

INTERNAL CONTROLS

Detailed internal control information is not available for this dataset; therefore, detection is based solely on observable transaction patterns.

SCENARIO-BASED DATA ANALYTICS

SCENARIO 1:

An external fraudster gains unauthorized access to an active customer account and executes multiple high-frequency transactions within a short time window to extract funds before detection.

Person: External fraudster

Entity: Customer account (real, non-complicit)

Action: Unauthorized access + rapid transaction execution

Impact: Loss of funds

Conversion: Transfer to external / mule accounts / Purchase of goods, services/ cash withdrawal

Concealment:

- Speed: Extract funds before detection systems react
- Fragmentation: Break large amounts into smaller transactions
- Normalisation: Keep amounts within “normal-looking” ranges

Red flags:

- Speed: High transaction frequency
- Speed: Low time between transactions
- Fragmentation: Multiple sub-threshold transactions where transaction amounts show low variation within the same day
- Normalisation: Abnormal cumulative amount

Designing search routine for scenario 1:

Relevant data fields:

TransactionID, AccountID, TransactionAmount, TransactionDateTime (recoded TransactionDate feature)

Behavioural baseline for user:

- Total transactions for AccountID
- Total transaction each date for AccountID
- Average daily number of transactions for AccountID -> The maximum number of transaction per AccountID per day is 3.
- Average daily transaction amount for AccountID

Detection strategy:

The date field only contains a timestamp for a limited number of transactions in the dataset, so it will not be possible to detect suspicious transactions based on the minutes since last transaction. Furthermore, the maximum number of transactions per AccountID per day is 3, which means that we will not find cases where a high number of transactions were executed rapidly. The approach to detecting Scenario 1 should be calibrated to the data that we have.

So, the following detection strategies are chosen:

- Identify days where transaction frequency is unusually high relative to the user baseline
- Identify days where cumulative transaction value significantly exceeds normal behaviour
- Detect patterns of multiple transactions with similar values indicating fragmentation
- Combine indicators to flag transactions consistent with rapid extraction behaviour.

Data interrogation tests:

Test 1: Volume anomaly

transaction_per_day > 1

Test 2: Amount anomaly

transaction_per_day >1
AND daily_total_amount > 1.5 * avg_daily_amount

Test 3: Fragmentation pattern

transaction_per_day >1
AND multiple transactions with similar amounts within same day

SCENARIO 2:

An external fraudster identifies and gains access to a dormant or inactive customer financial account and initiates transaction activity after a prolonged period of inactivity to extract funds, resulting in financial loss.

Person: External fraudster

Entity: Customer account: real, non-complicit

Action: Unauthorized access + transaction activity after dormancy/inactivity

Impact: Loss of funds

Conversion: Transfer to external / mule accounts / purchase of goods or services / cash withdrawal

Concealment

- **Dormancy exploitation:** Fraudster targets an account with little or no recent activity because unusual activity may be noticed later.
- **Delayed detection:** Account owner may not actively monitor the account.
- **Unauthorised access:** Fraudster attempts multiple logins to gain unauthorized access to the account prior to executing transactions.

Red flags

- **Dormancy exploitation & delayed detection:** Long gap between previous transaction and current transaction
- **Unauthorised access:** abnormal amount of login attempts

Designing search routine for scenario 2:

Identify relevant data fields:

TransactionID, AccountID, TransactionAmount, TransactionDateTime (recoded TransactionDate feature)

Establish behavioural baseline for user:

- Previous transaction date per account_id
- Days since previous transaction
- Average gap between transactions per account_id
- Maximum historical gap per account_id

- Average transaction amount per account_id

Detection strategy

Scenario 2 should identify accounts where transaction activity resumes after a prolonged inactivity period.

Chosen detection strategies:

- Identify transactions occurring after a long period of no prior activity.
- Compare the inactivity gap to the user's normal transaction gap.
- Identify whether the resumed activity involves abnormal transaction value.
- Flag transactions consistent with dormant account takeover behaviour.

Data interrogation tests:

Test 1: Dormancy / inactivity gap

$\text{days_since_previous_transaction} > 2 * \text{avg_days_between_transactions}$
AND $\text{days_since_previous_transaction} > 30$

Test 2: Amount anomaly

$\text{transaction_amount} > 1.5 * \text{avg_transaction_amount}$

Test 3: Suspicious login

$\text{login_attempts} > 2$

CUMULATIVE ANALYSIS

Each transaction is evaluated across all scenario-based tests. Individual red flags are aggregated to produce a cumulative risk score.

Scenario 1 and Scenario 2 flags are combined to identify transactions exhibiting multiple fraud indicators.

Particular emphasis is placed on transactions that:

- Occur after prolonged inactivity (Scenario 2)
- Exhibit abnormal transaction behaviour (Scenario 1)

These combinations represent higher-risk events consistent with unauthorized account takeover. Transactions flagged across multiple scenarios are prioritised, as the presence of multiple independent red flags increases the likelihood that the observed behaviour is consistent with a fraud scenario.

CONCLUSION

The analysis identified transactions exhibiting patterns consistent with unauthorized account usage.

Transactions flagged under Scenario 1 demonstrate abnormal transaction behaviour, including elevated transaction frequency and unusually high cumulative transaction amounts. Transactions flagged under Scenario 2 highlight accounts where activity resumed after prolonged inactivity, indicating potential dormant account exploitation.

Transactions flagged across both scenarios represent higher-risk cases, as they combine behavioural anomalies with temporal inactivity patterns. These cases are most consistent with account takeover fraud.

While the dataset lacks confirmation labels, the methodology successfully isolates transactions with an elevated likelihood of fraud based on scenario-driven red flag analysis.